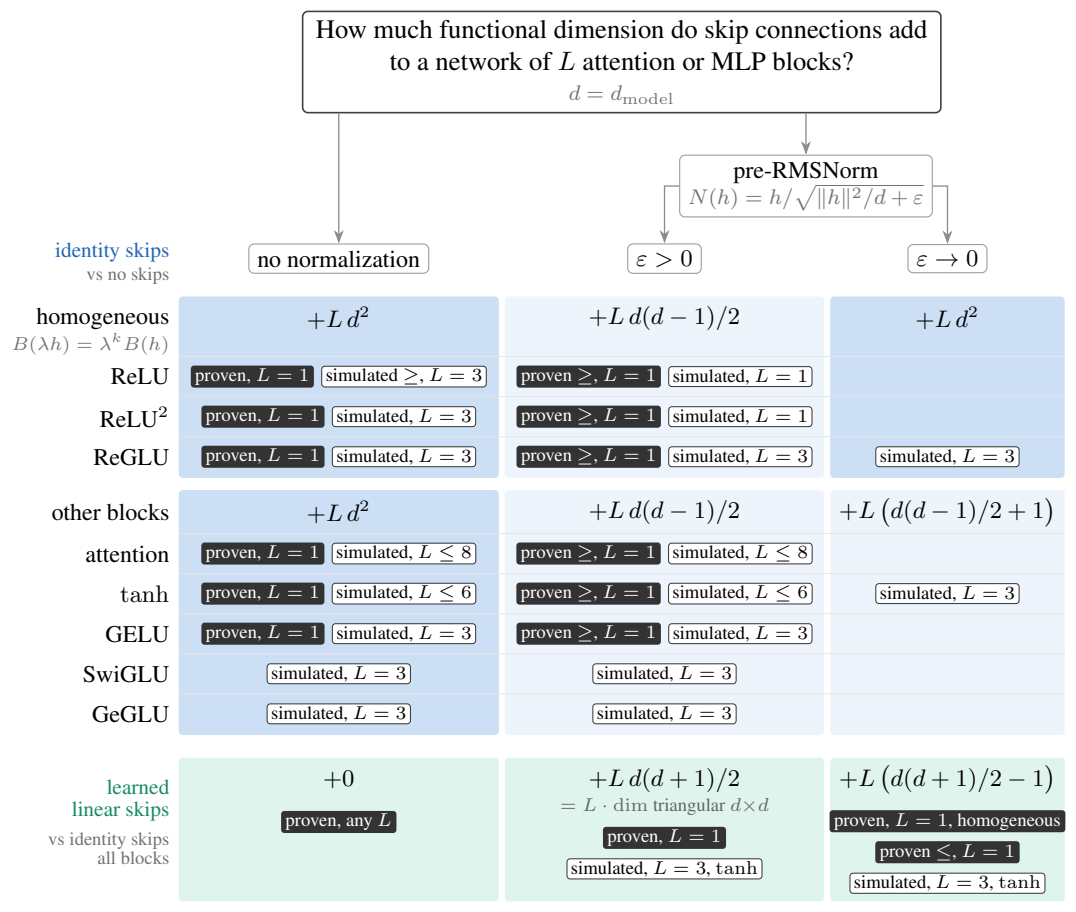

How much functional dimension do skip connections add to a network of L attention or MLP blocks?
$d = d_{\text{model}}$
pre-RMSNorm
$N(h) = h/\sqrt{\|h\|^2/d + \varepsilon}$
identity skips
vs no skips
no normalization
$\varepsilon > 0$
$\varepsilon \to 0$
homogeneous
$B(\lambda h) = \lambda^k B(h)$
$+L\,d^2$
$+L\,d(d-1)/2$
$+L\,d^2$
ReLU
proven, $L = 1$
simulated $\geq$, $L = 3$
proven $\geq$, $L = 1$
simulated, $L = 1$
ReLU^2
proven, $L = 1$
simulated, $L = 3$
proven $\geq$, $L = 1$
simulated, $L = 1$
ReGLU
proven, $L = 1$
simulated, $L = 3$
proven $\geq$, $L = 1$
simulated, $L = 3$
simulated, $L = 3$
other blocks
$+L\,d^2$
$+L\,d(d-1)/2$
$+L\left(d(d-1)/2+1\right)$
attention
proven, $L = 1$
simulated, $L \leq 8$
proven $\geq$, $L = 1$
simulated, $L \leq 8$
tanh
proven, $L = 1$
simulated, $L \leq 6$
proven $\geq$, $L = 1$
simulated, $L \leq 6$
simulated, $L = 3$
GELU
proven, $L = 1$
simulated, $L = 3$
proven $\geq$, $L = 1$
simulated, $L = 3$
SwiGLU
simulated, $L = 3$
simulated, $L = 3$
GeGLU
simulated, $L = 3$
simulated, $L = 3$
learned
linear skips
vs identity skips
all blocks
$+0$
proven, any L
$+L\,d(d+1)/2$
$= L \cdot \dim$ triangular $d \times d$
proven, $L = 1$
simulated, $L = 3$, tanh
$+L\left(d(d+1)/2-1\right)$
proven, $L = 1$, homogeneous
proven $\leq$, $L = 1$
simulated, $L = 3$, tanh